

# Smart Switch For AGRICULTURE IRRIGATION

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This smart switch system comprises an Arduino board with a GSM module that is remotely controlled by a smartphone. Through this adaptive technology, users can easily monitor the working condition and operational mode of machinery used on their farms from any location, rather than traveling long distances to the field under unfavourable conditions. The main objective is to provide convenience to the farmers.

Fig. 1 displays the authors' prototype on a breadboard. The components required to build this system are listed in the Bill of Materials table.

## Circuit and working

The circuit diagram of the smart switch for agricultural irrigation is shown in Fig. 2. As shown, the switch is constructed around a 50-watt solar panel, a solar charge controller, Arduino Uno board, GSM with GPRS Sim800l module, 16x2 LCD display, four-channel relay module, and a few other components.

Once the components are connected according to the circuit diagram, the entire device can be powered with the solar panel, and the DC-DC converter converts the voltage to the device's voltage range. The smart switch system operates using Arduino Uno, GSM module, and a smartphone.

The aim of this system is to control different agricultural machinery using a smartphone. When the power is turned on, the connection LED on the GSM module starts blinking. To initiate control, one needs to launch the app (developed through MIT app inventor) on the smartphone and connect to the GSM module. Upon

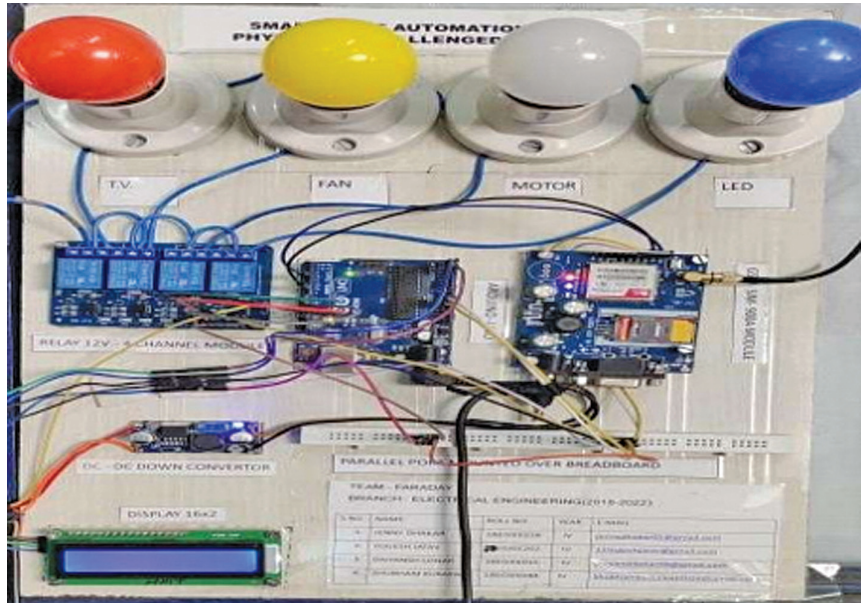


Fig. 1: Authors' prototype

successful pairing, the LED's blinking stabilises.

In the app, different keys are set to transmit different loads and their corresponding values. When a key is pressed on the smartphone, the GSM module receives the corresponding data and internally transmits it to Arduino. For instance, if 'motor on' is sent, the data is transmitted to Arduino, which compares it with the predefined data in the sketch and activates the 'motor on' accordingly. This action is applica-

## BILL OF MATERIALS

| Components                     | Description  | Quantity               |
|--------------------------------|--------------|------------------------|
| Arduino Uno/Nano (MOD1)        | 7V-12V       | 1                      |
| GSM with GPRS Sim800l (MOD2)   | 5V           | 1                      |
| Four-channel relay module      | 5V           | Depends on machineries |
| DC-DC down converter           | 2V to 35V    | 1                      |
| LCD display (16x2)             | 5V           | 1                      |
| Inter-integrated circuit (I2C) | 5V           | 1                      |
| Contactors                     | 200V to 600V | Depends on machineries |
| Connecting wires               | --           | ---                    |
| Solar panel                    | 50W          | 1                      |
| Solar charge controller        | 12V          | 1                      |

ble to other keys and loads as well.

The code is developed using the Arduino IDE. The Liquid Crystal I2C library is utilised in the code; one needs to install the library using the library manager. Then, configure the I2C display address and display size

## EFY Note

The source code of this project is available

for free download at [www.electronicsforu.com](http://www.electronicsforu.com)